

HARROW, ON, Canada

WMO: 712980

Lat: 42.033N

Lon: 82.896W

Elev: 631

StdP: 14.36

Time Zone: -5.00 (NAE)

Period: 00-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	2.7	7.8	-5.0	4.3	3.6	-0.2	5.6	9.0	24.2	21.8	22.3	22.6	7.9	250	0.483

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.7	87.3	73.3	84.9	72.1	82.4	70.8	76.2	83.2	74.6	81.3	73.1	79.8	9.1	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
74.1	130.3	80.5	72.4	122.9	78.9	70.8	116.3	77.5	40.1	83.2	38.5	81.3	37.2	79.9	82.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
20.8	18.3	16.1	-3.2	91.5	5.2	2.9	-7.0	93.6	-10.0	95.3	-13.0	96.9	-16.8	99.0		
			-3.9	78.8	5.0	2.3	-7.5	80.5	-10.5	81.9	-13.3	83.2	-16.9	84.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	49.7	25.3	27.1	35.6	47.2	58.7	68.5	72.5	70.7	64.4	52.9	41.4	30.6
	DBStd	18.65	10.71	9.61	9.62	8.51	8.26	6.58	5.59	5.13	7.13	8.45	8.40	9.13
	HDD50	2980	765	642	458	151	15	0	0	0	1	65	280	603
	HDD65	6286	1229	1062	911	537	233	38	7	11	97	385	709	1067
	CDD50	2868	0	1	12	66	285	555	699	640	432	156	21	1
	CDD65	699	0	0	0	2	38	143	241	186	78	11	0	0
	CDH74	4738	0	0	0	15	249	968	1854	1192	424	36	0	0
	CDH80	1095	0	0	0	0	46	261	483	219	82	4	0	0
Wind	WSAvg	7.3	9.3	8.6	8.2	8.3	6.8	6.2	5.5	5.2	5.6	7.2	8.3	8.9
Precipitation	PrecAvg	35.2	2.5	2.3	2.6	3.3	3.5	3.1	3.4	3.2	3.5	2.6	2.7	2.7
	PrecMax	55.6	4.1	5.3	5.6	5.5	6.7	5.3	6.5	6.6	7.3	5.7	7.0	5.4
	PrecMin	28.0	0.8	0.7	0.6	1.2	1.3	1.2	1.0	0.4	1.0	0.4	0.6	1.2
	PrecStd	5.6	0.9	1.1	1.1	1.1	1.7	1.3	1.4	1.5	1.9	1.2	1.3	0.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	52.3	56.5	67.1	76.3	84.7	89.5	90.8	88.5	86.7	78.4	65.0	55.7
		MCWB	50.5	52.0	59.4	62.7	69.5	72.3	74.9	74.6	73.3	68.6	57.0	52.6
	2%	DB	48.4	48.7	59.0	70.7	80.5	86.3	87.7	84.9	82.3	73.1	60.2	50.7
		MCWB	46.6	45.5	53.1	58.8	67.3	71.2	74.0	73.2	71.1	65.3	54.8	48.6
	5%	DB	43.3	43.3	54.6	65.5	76.5	83.1	85.2	82.3	78.7	69.5	57.0	46.2
		MCWB	40.9	40.1	49.9	56.0	65.6	69.9	72.6	71.8	68.8	62.4	52.1	43.6
	10%	DB	39.1	39.0	49.9	61.0	72.1	79.7	82.5	80.0	75.6	66.1	53.9	42.3
		MCWB	36.8	35.9	45.5	52.7	63.0	67.8	71.4	70.6	67.4	59.5	49.7	39.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	50.9	52.5	61.0	64.4	72.4	75.4	79.7	78.0	75.9	70.0	59.4	52.9
		MCDB	52.0	55.1	66.9	73.4	81.6	84.7	87.3	84.1	84.2	75.9	62.6	55.2
	2%	WB	46.8	45.8	54.3	60.7	69.8	73.5	76.8	75.6	73.0	67.0	55.6	48.7
		MCDB	48.2	48.5	57.8	68.0	76.9	82.7	83.6	81.6	79.0	71.5	59.3	50.5
	5%	WB	41.2	40.2	50.3	57.3	67.0	71.9	75.1	74.0	71.0	63.4	52.7	43.8
		MCDB	42.7	42.9	54.0	63.9	73.6	80.1	81.7	79.8	76.7	68.1	56.2	45.8
	10%	WB	37.0	36.2	45.6	53.9	64.5	70.3	73.5	72.4	69.2	60.3	49.9	40.2
		MCDB	38.9	38.7	49.5	60.1	71.1	77.4	80.0	78.1	74.2	65.1	53.4	42.1
Mean Daily Temperature Range	5% DB	MDBR	11.5	12.1	13.8	17.1	17.0	16.4	16.7	16.1	16.5	14.8	12.2	9.9
		MCDBR	13.8	16.1	19.7	22.8	20.2	18.8	17.8	17.1	18.0	17.6	15.5	12.8
		MCWBR	13.1	14.1	15.1	14.7	10.8	8.4	8.3	8.7	9.4	10.4	12.4	12.0
	5% WB	MCDBR	13.7	15.8	18.6	20.8	17.9	16.7	15.6	14.3	15.3	15.6	14.5	13.0
		MCWBR	13.6	14.6	15.2	15.0	11.3	8.7	8.5	8.3	9.1	10.5	12.7	12.8
Clear-Sky Solar Irradiance	taub	0.305	0.334	0.355	0.398	0.434	0.447	0.438	0.430	0.387	0.361	0.321	0.315	
	taud	2.401	2.299	2.327	2.255	2.201	2.214	2.263	2.274	2.366	2.419	2.493	2.425	
	Ebn at Noon	266	274	282	276	268	264	265	264	268	262	259	252	
	Edh at Noon	25	33	36	42	45	45	42	40	34	29	23	23	
All-Sky Solar Radiation	RadAvg	458	734	1102	1465	1720	1931	1933	1685	1360	886	553	384	
	RadStd	49	90	104	122	146	124	102	66	116	73	75	35	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (60 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+164	N/A

Nomenclature: See separate page